

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	BALANCED
Model	llama3.2

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Editorial Decision

MAJOR REVISION

Weighted Score: 3.4 / 5.0
Confidence: 78%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.2	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	0.5	Very Weak	■■■■■
Results & Evidence	4.0	Good	■■■■■
Discussion & Conclusions	4.0	Good	■■■■■
Figures & Tables	—	Skipped	—
Scientific Writing	—	Skipped	—
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.2 / 5.0 | Confidence: 90%

Weaknesses:

- [TITLE] could be shortened to fit within the recommended length.

Major Comments:

- [ABSTRACT] should not contain references as per standard format.

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 90%

Strengths:

- Clear research problem stated in Introduction
- Research question or hypothesis explicitly declared
- Objectives stated and later addressed in conclusions

Minor Comments:

- Transitions and coherence between sections could be improved

Recommendations:

- Consider adding more transitional phrases to improve flow between sections
- Ensure consistency in formatting throughout the manuscript

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The proposed approach integrates physical admissibility constraints into the neural mapping, decoupling the computationally intensive training phase from deployment.

Major Comments:

- The logical flow of the paper could be improved by explicitly stating the problem and its relevance to civil engineering applications earlier in the introduction.
- The experimental setup and dataset are not thoroughly described, which may impact the reproducibility of the results.

Recommendations:

- Consider adding more details about the experimental setup and dataset to improve reproducibility.
- Clarify the relevance of the proposed approach to civil engineering applications earlier in the introduction.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 0.5 / 5.0 | Confidence: 53%

Strengths:

- Bootstrap confidence intervals used with sufficient resamples
- RMSE, MAE, and R² metrics used and interpreted appropriately

Major Comments:

- The manuscript could benefit from a more detailed explanation of the assumptions verified for the Kruskal-Wallis test.
- It is unclear whether the sample size of 30 independent runs is sufficient to capture the underlying distribution of the data.

Recommendations:

- Verify normality of residuals in Kruskal-Wallis test using visual checks or statistical tests (e.g., Shapiro-Wilk test).
- Justify the sample size of 30 independent runs based on power analysis or simulation studies.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **0.0 / 5.0** | Confidence: 0%

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript clearly states the research objectives and presents a clear framework for the EANN approach.

Weaknesses:

- Some results, such as Pareto analysis and hypervolume/spread indicators, are not explicitly tied to figures or tables. It would be helpful to include these in the tables or figures for better clarity.

Major Comments:

- The manuscript provides a clear description of the experimental setup and validation process. However, it would be beneficial to provide more details on the preprocessing steps applied to the measured signals.

Recommendations:

- Include Pareto analysis and hypervolume/spread indicators in tables or figures for better clarity.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **4.0 / 5.0** | Confidence: 80%

Weaknesses:

- The discussion could benefit from more explicit comparison with prior work in the field of structural parameter identification.

Major Comments:

- The conclusions are directly derived from the results, but it would be beneficial to provide more context on how these findings contribute to the broader field of structural health monitoring.

Recommendations:

- Consider adding a section to discuss the implications of using EANN for near real-time integration into structural health monitoring pipelines and adaptive control architectures.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **0.0 / 5.0** | Confidence: 0%

Weaknesses:

- Could not parse model response — raw output stored

Recommendations:

- Re-run this agent or check model output

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 90%

Weaknesses:

- References are mostly from the last few years, but lack a broader range of seminal works and recent developments in the field.

Major Comments:

- The references section could benefit from more diverse sources, including more recent seminal works and studies outside of the author's research group.

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Include more recent seminal works to strengthen the manuscript's originality and novelty.
- Consider adding conference papers or grey literature to balance the references with journal articles.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [ABSTRACT] should not contain references as per standard format.
- [2] The logical flow of the paper could be improved by explicitly stating the problem and its relevance to civil engineering applications earlier in the introduction.
- [3] The experimental setup and dataset are not thoroughly described, which may impact the reproducibility of the results.
- [4] The manuscript could benefit from a more detailed explanation of the assumptions verified for the Kruskal-Wallis test.
- [5] It is unclear whether the sample size of 30 independent runs is sufficient to capture the underlying distribution of the data.
- [6] The manuscript provides a clear description of the experimental setup and validation process. However, it would be beneficial to provide more details on the preprocessing steps applied to the measured signals.
- [7] The conclusions are directly derived from the results, but it would be beneficial to provide more context on how these findings contribute to the broader field of structural health monitoring.
- [8] The references section could benefit from more diverse sources, including more recent seminal works and studies outside of the author's research group.
- [9] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] Transitions and coherence between sections could be improved
- [2] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. Consider adding more transitional phrases to improve flow between sections
2. Ensure consistency in formatting throughout the manuscript
3. Consider adding more details about the experimental setup and dataset to improve reproducibility.
4. Clarify the relevance of the proposed approach to civil engineering applications earlier in the introduction.
5. Verify normality of residuals in Kruskal-Wallis test using visual checks or statistical tests (e.g., Shapiro-Wilk test).
6. Justify the sample size of 30 independent runs based on power analysis or simulation studies.
7. Include Pareto analysis and hypervolume/spread indicators in tables or figures for better clarity.
8. Consider adding a section to discuss the implications of using EANN for near real-time integration into structural health monitoring pipelines and adaptive control architectures.
9. Re-run this agent or check model output
10. Include more recent seminal works to strengthen the manuscript's originality and novelty.
11. Consider adding conference papers or grey literature to balance the references with journal articles.
12. Include a conflict-of-interest declaration.